

Lecture 2: Majorana zero modes (cont.)

Monday, February 9, 2015 10:27 PM

More on the properties of Majoranas (e.g. braiding) soon. We will now change gears, and discuss Majoranas in *one dimensional* systems.

Here, we are mostly following Kitaev (2003).

Consider the following lattice model for a 1D

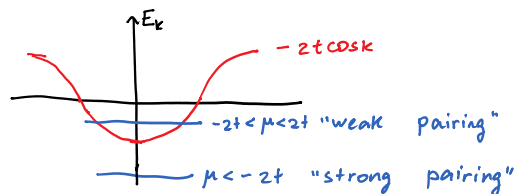
S.C.:

$$H = \sum_j \left(-t \psi_j^\dagger \psi_{j+1} - \Delta \psi_j^\dagger \psi_{j+1}^\dagger \right) + \text{h.c.} - \mu \sum_j \psi_j^\dagger \psi_j$$

This can be solved for a system with periodic boundary conditions, as before, by a Fourier transform.

The spectrum is given by

$$E_k = \pm \sqrt{(-2t \cos k - \mu)^2 + (2\Delta \sin k)^2}$$



Again, there is a gap closing (phase transition) between the weak and strong pairing phases.

What happens at the edge of the weak pairing phase?

Consider a finite system with N sites, at the special case $\Delta = t, \mu = 0$.

○ — ○ — ... — ○

Write every fermion ψ_j as two

Majorana operators, α_j, β_j :

$$\psi_j = \frac{1}{2} (\alpha_j + i\beta_j) \quad \alpha_j = (\psi_j + \psi_j^\dagger)$$

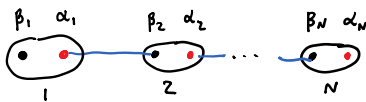
$$\psi_j^\dagger = \frac{1}{2} (\alpha_j - i\beta_j) \quad \beta_j = \frac{1}{i} (\psi_j - \psi_j^\dagger)$$

$$\{\alpha_i, \alpha_j\} = 2\delta_{ij}$$

Insert into H :

$$H = \sum_{j=1}^{N-1} -t \frac{1}{4} (\alpha_j + i\beta_j)(\alpha_{j+1} - i\beta_{j+1}) - \frac{t}{4} (\alpha_j - i\beta_j)(\alpha_{j+1} + i\beta_{j+1}) + h.c.$$

$$= \sum_{j=1}^N t i\alpha_j \beta_{j+1}$$

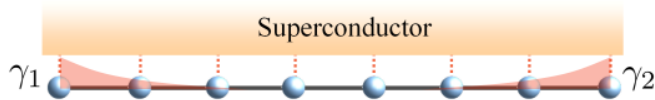


$$[\beta_1, H] = [\alpha_N, H] = 0 \Rightarrow \text{These are}$$

zero modes.

As before, the zero modes are robust

Lessons from the one dimensional topological superconductor



- Gapped system, two degenerate ground states, characterized by having a different fermion parity
- Defects (in this case, the edges of the system) carry protected zero modes described by anti-commuting operators: $\gamma_1 \gamma_2 = -\gamma_2 \gamma_1$
- Ground state degeneracy is "topological": no local measurement can distinguish between the two states!
- Useful as a "quantum bit"?

- Braiding of vortices / zero modes.

- Robustness to interactions, many-body interpretation in 1D.

- Fractionalized Majoranas.

In the presence of N Majorana zero modes, the g.s. manifold is $2^{N/2}$ fold degenerate.





Every pair of Majoranas form a 2-level system.

Basis: $i\gamma_{2j-1}\gamma_{2j} = \pm 1$

$\Leftrightarrow$ form a fermion:

$$f_j = \frac{1}{2} (\gamma_{2j-1} + i\gamma_{2j})$$

and then $f_j^\dagger f_j = 0, 1.$

What happens when we exchange ("braid") two Majoranas?



(See Kareljan's talk)

How do the operators $\gamma_{1,2}$ transform under braiding?

$$\gamma_1 \rightarrow \gamma'_1 = U_{12}^\dagger \gamma_1 U_{12} \quad U_{12}: \text{Unitary adiabatic evolution}$$

$$\gamma_2 \rightarrow \gamma'_2 = U_{12}^\dagger \gamma_2 U_{12} \quad \text{Operator.}$$

We expect

$$\gamma'_1 \propto \gamma_2 \quad (\text{up to a phase})$$

$$\gamma'_2 \propto \gamma_1$$

$$(\gamma'_1)^2 = U_{12}^\dagger \gamma_1 U_{12} U_{12}^\dagger \gamma_1 U_{12} = 1 \Rightarrow \text{phases are } \pm 1$$

suppose $\gamma'_1 = \gamma_2$

transformation has to conserve $i\gamma_1\gamma_2$ (fermion parity of 1,2)

$$\rightarrow \gamma'_2 = -\gamma_1!$$

One can check that the transformation that does this is

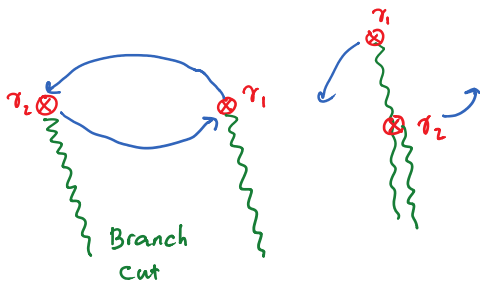
$$U_{12} = e^{i\phi} e^{\frac{\pi}{4}\gamma_1\gamma_2}$$

phase we can't determine from the present considerations.

check: $e^{\frac{\pi}{4}\gamma_1\gamma_2} = \frac{1}{\sqrt{2}} (1 + \gamma_1\gamma_2)$

$$\begin{aligned}
e^{-\frac{\pi}{4} \tau_1 \tau_2} \tau_1 e^{\frac{\pi}{4} \tau_1 \tau_2} &= \frac{1}{2} (1 - \tau_1 \tau_2) \tau_1 (1 + \tau_1 \tau_2) \\
&= \frac{1}{2} (\tau_1 + \tau_2 + \tau_2 - \tau_1) \\
&= \tau_2 \\
e^{-\frac{\pi}{4} \tau_1 \tau_2} \tau_2 e^{\frac{\pi}{4} \tau_1 \tau_2} &= \frac{1}{2} (1 - \tau_1 \tau_2) \tau_2 (1 + \tau_1 \tau_2) \\
&= \frac{1}{2} (\tau_2 - \tau_1 - \tau_1 - \tau_2) = -\tau_2
\end{aligned}$$

Ivanov (2001) argument: When a fermion goes around a vortex in a S.C., it gets a (-1) sign.



τ_2 collides with the branch cut of τ_2 once.

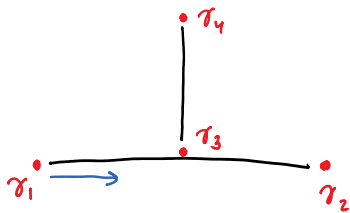
How to "braid" Majoranas on edges of wires?

(Alicea, Oreg, von Oppen, Refael, Fisher 10')

"exchange" τ_1, τ_2 ?



T-junction geometry: (two "auxiliary" Majoranas τ_3, τ_4 .)



Change position of τ_i (e.g., by applying gate potential(s)):

When τ_3 is close to τ_4 , $H_{\text{eff}} = i J_{34}(r_{34}) \tau_3 \tau_4$.

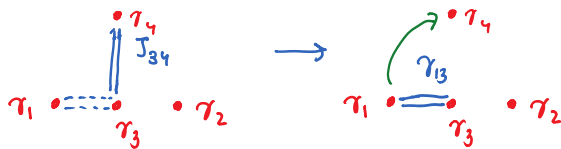
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More generally $H_{\text{eff}} = \sum_{i,j} i J_{ij}(t) \tau_i \tau_j$

Make the J_{ij} 's time dependent!

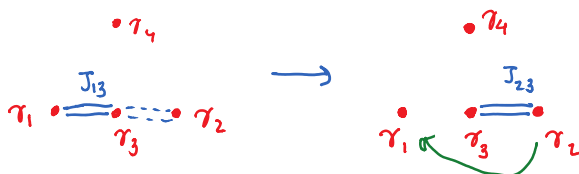
"Discrete Braiding protocol":

Step I:



(During this step, the g.s. degeneracy does not change!)

Step II:



Step III:



H returned to itself. However, the wavefunction does not.

E.g., in step I: $H_\lambda = -J(1-\lambda) i \tau_3 \tau_4 - J\lambda i \tau_1 \tau_3$

(τ_2 never participates)

λ goes from 0 to 1 adiabatically

Find zero mode of H_λ : $\gamma(\lambda) = \sum_{j=1,3,4} a_j \tau_j$

require $[\gamma(\lambda), H_\lambda] = 0$

$[\tau_1, \tau_1 \tau_3] = 2\tau_3$ etc.

$\Downarrow$

$$[\tau(\lambda), H] = a_1(-2\lambda J i \tau_3) + a_3(-2(1-\lambda)J \tau_4 + 2\lambda J \tau_1) + a_4(+2(1-\lambda)J i \tau_3) = 0$$

$$\begin{cases} -\lambda a_2 + (1-\lambda) a_4 = 0 \\ a_3 = 0 \end{cases}$$

$\Downarrow$

$$\lambda=0 : a_4 = 0, \quad a_2 = 1$$

$$\lambda=1 : a_2 = 0, \quad a_4 = +1 \text{ (you can find the sign by requiring that } a_4(\lambda) \text{ is continuous.)}$$

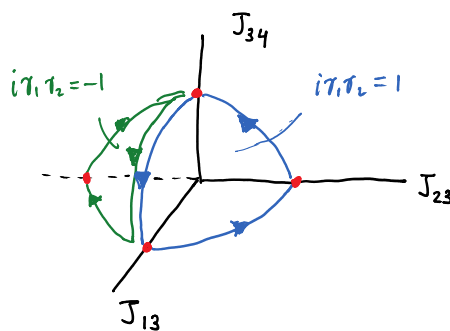
One can follow the procedure through steps II, III, and find that at the end $\begin{cases} \tau_1 \rightarrow \pm \tau_2 \\ \tau_2 \rightarrow \mp \tau_1. \end{cases}$ as anticipated.

What we have done here is drawn a closed loop in Hamiltonian space, rather than real space.

The braiding protocol can be visualized as follows.

The Hamiltonian has 3 parameters, J_{13} , J_{23} , and J_{34} .

Draw them in a 3D space and fix $J_{13}^2 + J_{23}^2 + J_{34}^2 = J^2 = \text{const.}$



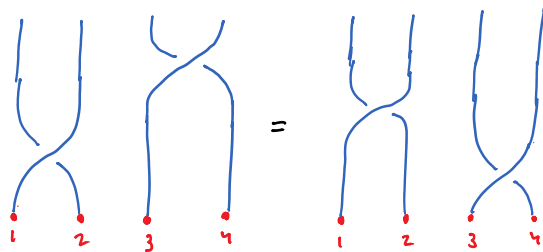
At any given intermediate time, there are only two of the three parameters which are non-zero. The trajectory covers $\frac{1}{8}$ of the area of the sphere; For a spin $\frac{1}{2}$ (2-level system), that gives a Berry phase of $\phi = \frac{1}{2} \cdot 4\pi \cdot \frac{1}{8} = \frac{\pi}{4}$. For the other eigenstate of $i\tau_1\tau_2$, we get the opposite phase.

Hence

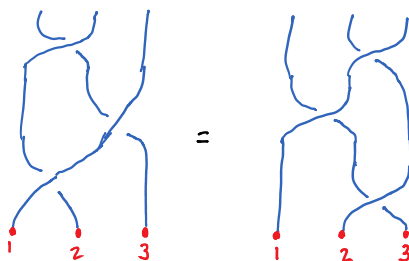
$$U_{12} = \begin{pmatrix} e^{i\pi/4} & 0 \\ 0 & e^{-i\pi/4} \end{pmatrix} = e^{-\frac{\pi}{4} \tau_1 \tau_2}$$

Notes:

- The result of the braiding protocol is topological; it does not depend on deviations from the path in Hamiltonian space, $\alpha J_{ij}(\lambda)$, as long as γ_2 remains decoupled in step 1, etc.
- The resulting braiding matrices U_{ij} form a projective representation of the braid group. By "projective" here we mean that the group relations are obeyed up to a phase.



$$U_{12} U_{34} = U_{34} U_{12}$$



$$U_{12} U_{23} U_{12} = U_{23} U_{12} U_{23} \quad (\text{Yang - Baxter equation})$$

- As mentioned earlier in this school, the Majorana (Ising anyon) braiding rules are non-universal for quantum computation purposes. This can be seen, e.g., from taking a single "q-bit" made out of four zero modes, and consider all the possible braiding operations.

To see this take e.g. $N=4$ zero modes, $\tau_1 \dots \tau_4$.

$\tau_1 \circ \tau_2$ fix the total fermion parity, $P = \tau_1 \tau_2 \tau_3 \tau_4$.

$\tau_3 \circ \tau_4$ The g.s. is now two fold degenerate.

e.g. it is spanned by $i\tau, \tau_z = \pm 1$;
 you can check that on this basis
 the following representation is possible

$$i\tau_1\tau_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma^z$$

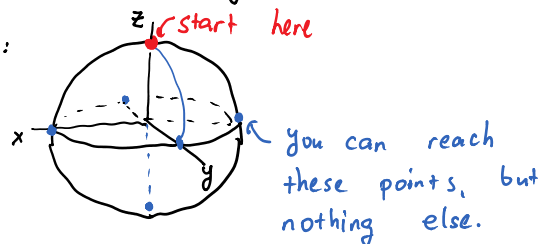
$$i\tau_1\tau_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \sigma^x$$

$$i\tau_1\tau_4 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \sigma^y$$

I.e. these form a "Pauli group".

The group is preserved by the braiding
 (meaning that every element is mapped to another element, but never
 a sum of two).

The braiding can be thought of as rotations by $\frac{\pi}{2}$
 around x, y, z :



This result generalizes to an arbitrary number of qbits.
 One can show that if the braiding preserves the
 a Pauli group defined on an n -qbit Hilbert space,
 then it is not universal.

Can we go beyond Majorana zero modes in one spatial dimension?

Suppose we allow for arbitrary interactions (beyond mean field theory). Could we get a phase with zero modes that extend the behavior of Majorana zero modes, and obtain the missing gates for universal TQC?

The answer to this question is believed to be no. To shed light on it, it is useful to understand the topological phase in the wire and the nature of Majoranas in one dimension in a more general way.

The Majorana chain can be mapped to the transverse field Ising model:

$$\Psi_j = (\prod_{k < j} \sigma_k^z) \sigma_j^-$$

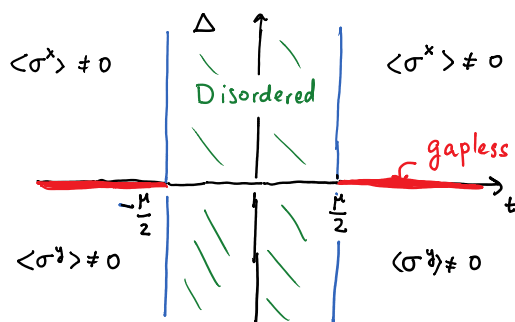
$$\Psi_j^\dagger \Psi_j = \sigma_j^+ \sigma_j^- = \frac{1 + \sigma_j^z}{2}$$

In terms of these variables

$$\begin{aligned} H &= -t \sum_j \Psi_j^\dagger \Psi_{j+1} - \Delta \sum_j \Psi_j^\dagger \Psi_{j+1}^\dagger + h.c. - \mu \sum_j \Psi_j^\dagger \Psi_j \\ &= -t \sum_j \sigma_j^+ \sigma_j^z \sigma_{j+1}^- - \Delta \sum_j \sigma_j^+ \sigma_j^z \sigma_{j+1}^+ + h.c. - \mu \sum_j \frac{1 + \sigma_j^z}{2} \\ &= -t \sum_j \sigma_j^+ \sigma_{j+1}^- - \Delta \sum_j \sigma_j^+ \sigma_{j+1}^+ + h.c. - \frac{\mu}{2} \sum_j \sigma_j^z + \text{const.} \\ &= -t \sum_j \frac{\sigma_j^x + i \sigma_j^y}{2} \frac{\sigma_{j+1}^x + i \sigma_{j+1}^y}{2} + h.c. - \Delta \sum_j \frac{\sigma_j^x + i \sigma_j^y}{2} \frac{\sigma_{j+1}^x - i \sigma_{j+1}^y}{2} + h.c. \\ &\quad - \frac{\mu}{2} \sum_j \sigma_j^z \\ &= -\frac{t}{2} \sum_j (\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y) - \frac{\Delta}{2} \sum_j \sigma_j^x \sigma_{j+1}^x - \sigma_j^y \sigma_{j+1}^y - \frac{\mu}{2} \sum_j \sigma_j^z \\ &= -\sum_j \left(\frac{t+\Delta}{2} \sigma_j^x \sigma_{j+1}^x + \frac{t-\Delta}{2} \sigma_j^y \sigma_{j+1}^y \right) - \frac{\mu}{2} \sum_j \sigma_j^z \end{aligned}$$

This is the well-known XYZ model, which can have three gapped phases: disordered, ordered with $\langle \sigma^x \rangle \neq 0$, and ordered with $\langle \sigma^y \rangle \neq 0$.

Schematic phase diagram:



← This can be read from the spectrum:

$$E_k = \pm \sqrt{(-2t \cos k - \mu)^2 + (2\Delta \sin k)^2}$$

by locating the regions where the gap closes.

(For $t = \pm \Delta$, this is the usual transverse field Ising model.)

Note the symmetry that protects the $\langle \sigma^x \rangle \neq 0$ and $\langle \sigma^y \rangle \neq 0$ phases:

This is

$$[P, H] = 0 \quad \text{with} \quad P = \prod_i \sigma_i^z = \prod_i e^{i\pi(1-\psi_i^\dagger \psi_i)} = \text{fermion parity.}$$

A field in the x or y direction is a non-local operator in terms of the fermions.

This would still be true in the presence of arbitrary interactions between fermions.

There are general arguments (Turner, Pollmann, EB (2010); Fidkowski, Kitaev (2010); Schuch et al. (2011); Chen, Gu, Wen (2011)) that this is the end of the story in 1D; in an arbitrary interacting system of fermions, there could be only two stable gapped phases, the topological phase with Majoranas at the end and a trivial phase. If extra symmetries are present, more phases - symmetry protected topological (SPT) phases - are possible. However, as long as our system is made out of fermions, this won't give any new non-Abelian properties beyond those of Majoranas. To find something richer, our basic degrees of freedom need to be anyons, as we will see next.

Gapped phases of fermions

A. Turner, F. Pollmann, EB (2010)

No symmetries: only one gapped phase in 1D

(N. Shuch et al., 2011)

- Conservation of **fermion parity** with $Q = (-1)^{N_{\text{total}}}$



"Fractionalization" of the parity operator (in low-energy subspace) $Q = Q_A Q_B$

Q_A, Q_B either fermionic or bosonic!

$$Q^A Q^B = e^{i\mu} Q^B Q^A$$

➡ two distinct phases with $\mu = 0, \pi$